3 Fig. 3.1 shows the cross-section of a barrage built across a tidal bay. The barrage is part of a tidal power station.

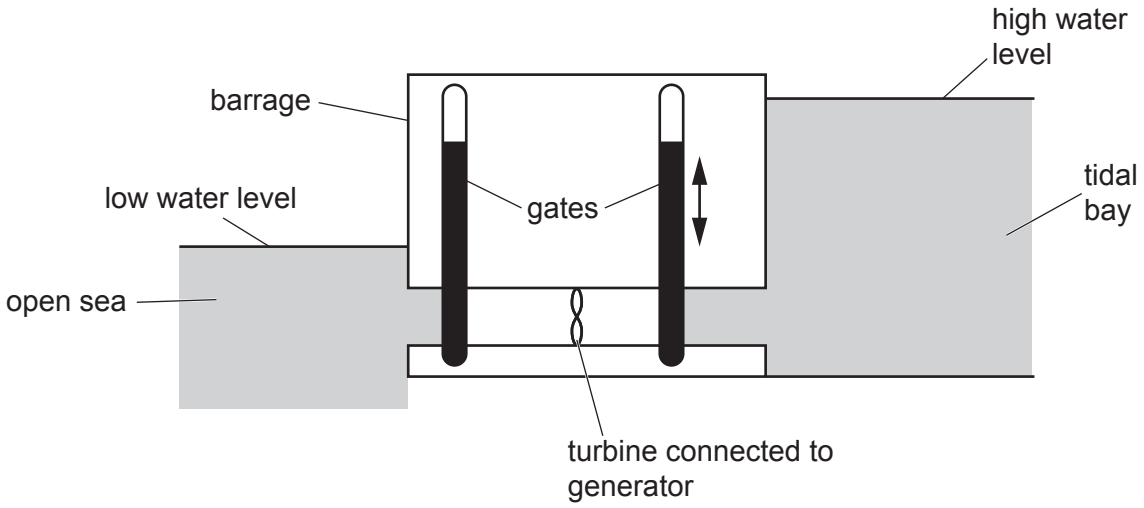


Fig. 3.1

The gates are raised to be open when the tide comes in. The gates are lowered to close when it is high tide. Fig. 3.1 shows the water levels in the open sea and the tidal bay when it is low tide. The gates are raised and water flows through the turbine.

(a) Complete the sentences to describe the energy transfers which take place when the gates are opened.

Use words from the list.

tidal bay **kinetic** **gates** **gravitational potential**
open sea **turbines** **water**

..... energy of the ..... in the

..... is transferred to ..... energy in the

rotating ..... . This energy is used in the generator to produce

electrical power. [3]

(b) State **one** advantage and **one** disadvantage of tidal power as an energy resource.

advantage .....

disadvantage .....
[2]

(c) State the **main** source of energy for tidal energy.

..... [1]

[Total: 6]